CPIS – 240

Database Midterm Exam

SECTION 1

A: Answer all questions by choosing one answer

1) ………… are responsible for identifying the data to be stored in the database and for choosing appropriate structures to represent and store this data.

A. Database Designers
B. End Users
C. Systems Analysts
D. Database Administrator

2) Degree is number of……….. in a relation

A. Entities
B. attributes
C. rows
D. columns

3) Whish of the following is NOT disadvantage of file based systems

A. Separation and isolation
B. Data dependence
C. Complexity
D. Redundancy of data

4) An ……………… is a structure that makes the search for particular database records efficient. It describes how records in a database file are retrieved. An access path may be either

A. Access path
B. Database state
C. Schema
D. valid state

5) What is the level that describe the part of the database that a particular user group is interested in and hides the rest of the database from that user group?

A. Internal level
B. External level
C. Logical Level
D. Physical Level

B: Match each letter with its corresponding term

1. The database structure called ……………**A**…………. and it changes ………………**B**………………

2. ………**C**…………. that is, a state that satisfies the structure and constraints specified in the schema

3. At any point in time, the database has ……………**D**………….

4. ……………**E**……………… is the set of allowable values for one or more attributes.

5. A bottom-up design process – combine a number of entity sets that share the same features into a higher-level entity set…………………**F**…………………….

6. Data model is a collection of concept that describe the structure of a database

……………**G**……………., relationship, and ……….**H**……….

Letter	The correct term
	Relationship
	Specialization
	Schema
	Attribute
	Data type
	Current state
	Empty state
	Initial state
	Domain
	Valid state
	Cardinality
	Access path
	Generalization
	Constrain
	Infrequently

SECTION 2

A: Based on the database instance below: Give a short answer to the following question

A course may enroll in by many students, and student may enroll many courses.

Course:

Course_ID	Major	Duration	Email
ACCT	Accounting	3	FEA@kau.edu.sa
CS	Computer science	4	fcit@kau.edu.sa
IS	Information system	4	fcit@kau.edu.sa
STAT	Statistics	3	scince@kau.edu.sa

Student:

Studen_ID	Student_Name	Course_ID	Email
2293639	Nada	CS	nada@kau.edu.sa
228374	Sara	STAT	sara@kau.edu.sa
227925	Reem	IS	reem@kau.edu.sa
226753	Nora	STAT	nora@kau.edu.sa
223468	Ameera	ACCT	ameera@kau.edu.sa

A. Candidate key for course?

....

B. Primary key for course?

……………………………………………………………………..

C. Foreign key for relation between course and student?

....

D. Cardinality for students?

………………………………………………………………………

B: **Based on the database instance below that represents the Suppliers (S) entity, Products (P) entity and the relationship between the two entities supplies (SP), would you accept or reject the following operations and specify the type of violation in case the operation is rejected.**

S:

SNO	SNAME	STATUS	CITY
S1	Smith	20	London
S2	Jones	10	Paris
S3	Blake	30	London
S4	Clark	20	Athens
S5	Adams	30	

SP:

SNO	PNO	QTY
S1	P1	300
S1	P2	200
S1	P3	400
S1	P4	200
S1	P5	100
S1	P6	100
S2	P1	300
S2	P2	400
S3	P2	200
S4	P2	200
S4	P4	300
S4	P5	400

P:

PNO	PNAME	COLOR	WEIGHT	CITY
P1	Nut	Red	12.0	London
P2	Bolt	Green	17.0	Paris
P3	Screw	Blue	17.0	Oslo
P4	Screw	Red	14.0	London
P5	Can	Blue	12.0	Paris
P6	Cog	Red	19.0	London

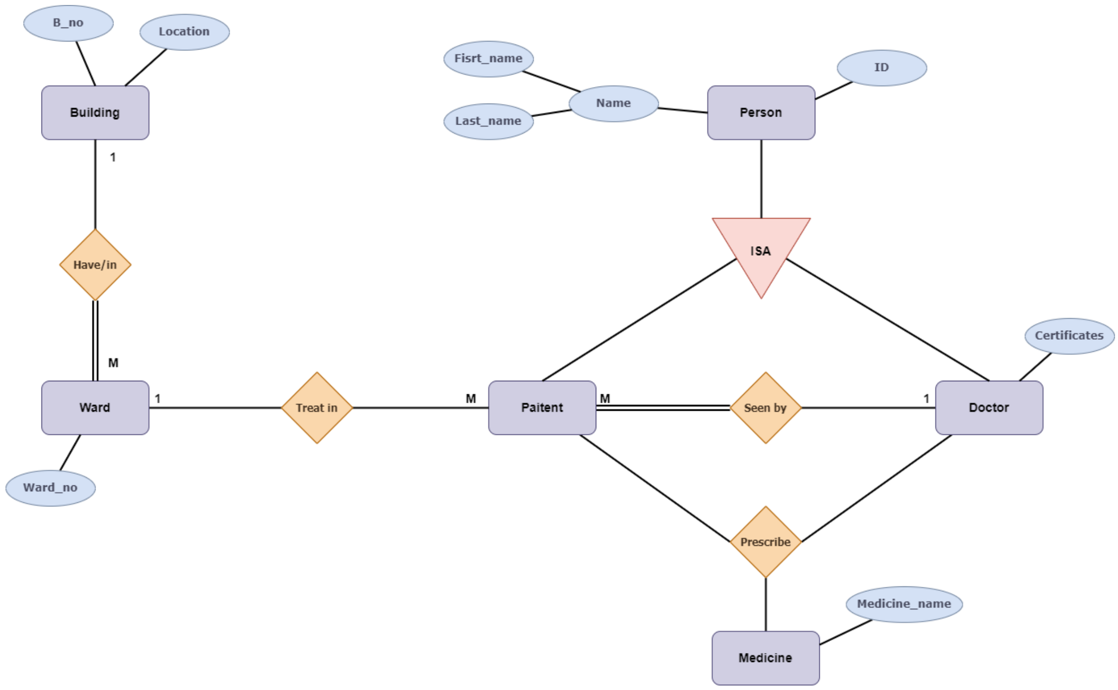
A. Delete from SP tuple where SNO = 'S3'

B. Insert <'S8, 'Ali', 'ZERO', 'Jeddah'> into S

C. Update table P Set PNO='P7' where PNO='P1'

SECTION 3: Given a partial ER-Diagram

A ward has number (1,2,3…), building has uniquely identified number, certification (شهادة) can be Saudi board Arab board, Canadian board …. Etc. Doctor can hold multiple certifications for example Saudi board and Canadian board.



Q1: The ER diagram above is incomplete, specify the following:

a. Weak entity ……………….. (show it in the diagram)

b. Multivalued ………………. (show it in the diagram)

c. In generalization, doctor can be a patient, and a patient can be a doctor

is that (Disjoin / overlapping) ?.......................

d. Patient ……………... (can / must) seen by (one / many) …………. doctor(s).

e. Show participation

Building (…… , ……) has (…… , ……) ward(s).

Q2: Write all entities that are participate in ternary relationship.

………………………………………………………………………………

SECTION 4: Draw an entity relationship diagram to show an information system for an estate company. Show all participation constraints that are described in the scenario.

Note: required (entities, attributes, relationships, minimum and maximum participation notations, total and partial relationships)

The company have several sales officers.
Each sale officer has an address, phone number, and unique id. A sale officer must have many employees, and sale officer should assign to many properties.
Each employee has a name and unique id. An employee may work for many sales officers.
Each property is described by an address and unique id. A property must assign to only one sales officers, and property may have many owners.
Each owner has a name and unique id. An owner may own many properties.

All property and owner relationship "Own" have percentage of the property that will be stored in the attribute "PercentProperty".

Good Luck

وفقكم الله ودعواتكم لنا ♥